Ecological Modelling 265 (2013) 114–123

Contents lists available at SciVerse ScienceDirect

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel

Time-dependent sensitivity of a process-based ecological model

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ARTICLE INFO

Article history:
Received 31 December 2012
Received in revised form 5 June 2013
Accepted 10 June 2013
Available online 7 July 2013

Keywords:
3-PG
Fourier amplitude sensitivity test
Process-based model
Temporal
Sensitivity analysis
Forest growth model

ABSTRACT

Sensitivity analysis is useful for understanding the behaviour of process-based ecological models. Often, time influences many model processes. Hence, the sensitivity of model outputs to variation in input parameters may also change with simulation period. We assessed the time-dependence of parameter sensitivity in a well-established forest growth model 3-PG (Physiological Principles for Predicting Growth) (Landsberg and Waring, 1997) as a case study. We used a screening method to select influential parameters for two key model outputs, i.e., stand volume and foliage biomass, then applied the Fourier amplitude sensitivity test (FAST) to quantify the sensitivity of the outputs to these selected parameters. Sensitivities were assessed on an annual time-step spanning 5–50 years of forest stand age. The influence of climatic and soil variables on time-dependent sensitivities was also quantified. We found that the sensitivities of most parameters changed substantially with forest stand age. Different climate and soil data also influenced the sensitivities of some parameters. Time-dependent sensitivity analysis provided much greater insight into model structure and behaviour than previous snapshot sensitivity analyses. Failing to account for time-dependence in sensitivity analysis could lead to misguided efforts in model calibration and parameter refinement, and the mis-identification of insensitive parameters for default value allocation. We concluded that sensitivity analysis should be conducted at simulation periods compatible with the process of interest. A more comprehensive sensitivity analysis scheme is required for temporal models to explore parameter sensitivities over the full simulation period and over the full variation in forcing data.

1. Introduction

Uncertainty analysis is an important part of the successful development, calibration, and application of process-based ecological models (Vrugt and Robinson, 2007). Due to the complex nature of ecological systems and the limited knowledge of the underlying processes, uncertainty is usually unavoidable. However, by identifying those parameters and processes most influential on model outputs through sensitivity analysis (SA), efforts could be guided towards improving the accuracies of the most influential parameters and be used to better understand model structure and behaviour, and reduce model uncertainty (Marino et al., 2008; Makler-Pick et al., 2011; Song et al., 2012). This is especially important for complex, process-based ecological models which can be richly parameterized (Wang et al., 2009).

Process-based ecological models often include time-dependent, non-linear processes (Landsberg and Waring, 1997; Thornton et al., 2002). This suggests that the contributions of each parameter to the variation in model outputs may also change with time. For forest growth models for example, the coupled non-linear reduction in stomatal conductance and hydraulic conductivity as trees age will inevitably influence all related physiological processes, e.g. photosynthesis and biomass allocation (Landsberg and Waring, 1997; Ryan and Yoder, 1997). Thereby, some parameters influential at young stand ages may decline in influence in older stands, and vice versa (Song et al., 2012). A time-dependent sensitivity analysis, which analyses model behaviour over the full simulation period horizon, is necessary for providing a more comprehensive understanding of the model structure, and to assist model calibration. However, many studies have undertaken

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http://dx.doi.org/10.1016/j.ecolmodel.2013.06.013

SA only at a single simulation period (e.g. Lu and Mohanty, 2001; Esprey et al., 2004; Makler-Pick et al., 2011). These types of *snapshot* SA may give a biased or unrepresentative view of parameter sensitivities corresponding to certain model outputs. Song et al. (2012) and Makler-Pick et al. (2011) suggested that time-dependence may change parameter sensitivities and highlighted this area for further research. A few other studies have also recognized this issue but have fallen short of quantifying the implications for temporal process-based ecological models (Cariboni et al., 2007; Marino et al., 2008; Xu et al., 2009). Further, model forcing data (e.g. climate and soil texture) may also influence parameter sensitivities (Fuentes et al., 2006; Almeida et al., 2007a). Ignoring the influence of forcing data may produce misleading results, especially when the environmental factors differ between sites. The influence of climate data is of great interest in this perspective as climate is likely to change over time.

Two types of SA are often used – local and global. Local SAs estimate the sensitivity of one parameter while holding all other parameters at central values. This provides an accurate estimate of parameter sensitivity under a very specific model condition and is only suitable for simple linear or additive models (Saltelli et al., 2008). For complex and parameter-rich models, global SAs are required which explore the full parameter space. Global SAs can be used to quantify the contribution of each parameter to the variation in the corresponding model output, and are useful for characterizing model structure and behaviour (Saltelli, 2000; Helton, 2008). Methods for global SA include the Fourier amplitude sensitivity test (FAST) (Cukier et al., 1973; Xu and Gertner, 2007), variance-based methods such as Sobol' (Sobol', 1990) and Saltelli's (Saltelli et al., 2010), and moment-independent method (Borgonovo, 2007). Variance-based methods can quantify both the main effect – the partial variance contributed by a certain parameter to the total variance of corresponding model output – and the interaction effects between parameters, but they are computationally intensive. Moment-independent method (Borgonovo, 2007) can also capture dependence among model parameters and consider sensitivity across the full distribution of model outputs, and are computationally efficient (Plischke et al., 2013). FAST is also computationally efficient (Saltelli et al., 1999; Xu and Gertner, 2011) and has been widely applied in analysing sensitivities in a wide range of models, including models of ecological (Xu et al., 2009), chemical (Lu and Mohanty, 2001), biological (Thogmartin, 2010), and atmospheric (Kioutsioukis et al., 2004).

The objectives of this study were to quantify the time-dependent parameter sensitivities for selected model outputs, and to demonstrate the significance of time-dependence in understanding model behaviour. A well known forest growth model, Physiological Principles for Predicting Growth (3-PG) (Landsberg and Waring, 1997) was used for this purpose. We first applied the computationally efficient Morris method to screen out the least influential parameters for each model output. We then applied the FAST method to analyse the sensitivities of the remaining parameters. Sensitivities were calculated at annual step from 5 to 50 years. The agreement among parameter sensitivities for each model output over time was quantified by computing the correlation coefficients on the ranking using Savage scores (Iman and Conover, 1987), and graphically visualized. We also analysed the influence of forcing data (e.g. climate and soil texture) on the time-dependence of parameter sensitivities. We discuss the impact of time-dependence on the effectiveness of sensitivity analysis in enabling the understanding of model structure and behaviour, and in informing model calibration. We also discuss the interrelationships among parameter distributions, sensitivities, and model output uncertainties.

2. Methods and data

2.1. Model description

Applications of process-based forest productivity models have increased recently due to their ability to simulate underlying physiological processes and to address questions in a range of fields such as forestry production (Almeida et al., 2010), carbon sequestration (Bryan and Crossman, 2013), biodiversity conservation (Crossman et al., 2011), bioenergy (Bryan et al., 2010), and food security (Paterson and Bryan, 2012). The 3-PG model is a simple process-based and stand-level deterministic forest growth model which captures the basic physiological principles of forest growth (Coops et al., 1998; Landsberg et al., 2001). 3-PG simulates growth in even-aged, homogeneous forests or plantations, and has been applied to a range of species including *Eucalyptus* species (Sands and Landsberg, 2002; Paul et al., 2007; Almeida et al., 2010), loblolly pine (Landsberg et al., 2001), and Douglas-Fir (Coops et al., 2010).

The 3-PG model has been continuously improved since its first publication in 1997. A detailed description of 3-PG can be found in Landsberg and Waring (1997), Sands and Landsberg (2002), and Paul et al. (2007). Important versions include 3-PGpjs (Sands and Landsberg, 2002; Sands, 2004), 3PG-Spatial (Tickle et al., 2001; Coops et al., 2010), and 3-PG2 (Almeida et al., 2007a). The latter includes a more detailed water balance calculations in forested landscapes. In this study, we used 3-PG2 and ported this Visual Basic for Applications (VBA) model into the Python programming language (http://python.org) to enable SA.

The 3-PG2 model can be used to predict a range of outputs, which include forest gross primary production (GPP), net primary production (NPP), foliage and stem biomass, water balance in forest soils and evapotranspiration. The model runs on a monthly time step. The input data required includes monthly average climate data (short wave solar radiation, mean maximum and minimum air temperature, vapour pressure deficit (VPD), frost days, rainfall and number of rainfall days), and site-specific parameters including latitude, soil texture, fertility rating (*FR*) (hereafter model parameters are represented as italic letters and model outputs in regular font), and initial number of stems per hectare. *FR* describes soil nutrient status as a score between 0 (for low fertility) and 1 (for high fertility) (Almeida et al., 2007b). Soil texture is classified into 12 classes. There are seven common soil attributes under each class to describe soil water related properties (Almeida et al., 2007a). Whilst 3-PG2 has more than sixty parameters, we selected 38 species-specific parameters for sensitivity analysis, excluding those parameters which have constant values (e.g. ratio of NPP and GPP, and molecular weight of dry matter) (Esprey et al., 2004; Almeida et al., 2007a; Song et al., 2012).

2.2. Data sources

We used one plot named Greenvale of *Corymbia maculata* and *Eucalyptus cladocalyx* located in Queensland, Australia, to demonstrate the time-dependent sensitivity properties of parameters in 3-PG2. This plot has a long climate record from 1930 to 2006, a *FR* of 0.2, and a clay soil texture. Two important biomass-related output variables, namely StandVol (stand volume excluding bark and branch ($m^3\ ha^{-1}$)) and WF (foliage biomass ($t\ ha^{-1}$)), were used in the following analysis.

Due to a lack of information about the prior probability distributions for each parameter, we assumed an independent uniform distribution for each parameter with bounds set at 30% either side of its reference value (Esprey et al., 2004; van Oijen et al., 2005; Song et al., 2012). Without loss of generality, the

Table 1
Parameters in 3-PG2 for sensitivity analysis and corresponding reference values for *C. maculata*/*E. cladocalyx*.

Parameter	Symbol	Unit	Reference value
Foliage:stem partitioning ratio @ DBH = 20 cm	*pFS20*	–	0.3
Constant in the stem mass vs. diam. Relationship	*aWS*	–	0.1032
Power in the stem mass vs. diam. Relationship	*nWS*	–	2.5447
Maximum fraction of NPP to roots	*pRx*	–	0.8
Minimum fraction of NPP to roots	*pRn*	–	0.2
Volume of soil accessed by 1 kg of root dry matter	*spRootVol*	$m^3\ kg^{-1}$	3.8
Maximum litterfall rate	*gammaF1*	$month^{-1}$	0.008
Litterfall rate at $t = 0$	*gammaF0*	$month^{-1}$	0.001
Age at which litterfall rate has median value	*tgammaF*	months	10
Average monthly root turnover rate	*gammaR*	$month^{-1}$	0.015
Minimum temperature for growth	T_{min}	°C	10
Optimum temperature for growth	T_{opt}	°C	30
Maximum temperature for growth	T_{max}	°C	40
Value of 'fNutr' when fertility rating = 0	*fN0*	–	0.6
Maximum stand age used in age modifier	*MaxAge*	years	50
Power in age modifier	*nAge*	–	4
Relative age at age modifier = 0.5	*rAge*	–	0.95
Max. stem mass per tree @ 1000 trees/hectare	*wSx1000*	$kg\ tree^{-1}$	300
Specific leaf area at age 0	*SLA0*	$m^2\ kg^{-1}$	5.365
Specific leaf area for mature leaves	*SLA1*	$m^2\ kg^{-1}$	5.365
Age at which specific leaf area = (*SLA0* + *SLA1*)/2	*tSLA*	years	2.5
Extinction coefficient for absorption of PAR by canopy	*k*	–	0.5
Age at canopy cover	*fullCanAge*	years	3
LAI for 50% reduction of VPD in canopy	*cVPD0*		5
Maximum thickness of water retained on leaves	*tWaterMax*	mm	0.25
Maximum proportion of rainfall evaporated from canopy	*MaxIntcptn*	–	0.15
LAI for maximum rainfall interception	*LAImaxIntcptn*	–	3
Maximum canopy quantum efficiency	*alphaCx*	$mol\ mol^{-1}$	0.06
Maximum canopy conductance	*MaxCond*	$m\ s^{-1}$	0.02
LAI for maximum canopy conductance	*LAIgcx*	–	3.33
Defines stomatal response to VPD	*CoeffCond*	$mbar^{-1}$	0.05
Canopy aerodynamic conductance	*gAc*	$m\ s^{-1}$	0.22
Branch and bark fraction at age 0	*fracBB0*	–	0.6
Branch and bark fraction for mature stands	*fracBB1*	–	0.41
Age at which *fracBB* = (*fracBB0* + *fracBB1*)/2	*tBB*	years	4.3
Minimum basic density – for young trees	*rho0*	$t\ m^{-3}$	0.6
Maximum basic density – for older trees	*rho1*	$t\ m^{-3}$	0.6
Age at which whole tree basic density = (*rhoMin* + *rhoMax*)/2	*tRho*	years	4.5

FR was assumed to follow a uniform distribution between 0.2 and 0.9.

2.3. Parameter screening

We used the elementary effects (EE) screening method proposed by Morris (1991) and improved by Campolongo et al. (2007) to screen the parameters listed in Table 1. The EE method is computationally efficient in screening out the least influential parameters in a model, and is especially useful in SA for parameter-rich models (Lu and Mohanty, 2001). A brief description of the EE method is as follows.

Given a model *f*(*X*), in which $X = (x_1, \ldots, x_{i-1}, x_i, \ldots, x_k)$ represents one point randomly sampled from the parameter space. The elementary effect of the *i*th parameter is defined as:

$$EE_i(X) = \frac{f(x_1, \ldots, x_{i-1}, x_i + \Delta, \ldots, x_k) - f(x_1, \ldots, x_{i-1}, x_i, \ldots, x_k)}{\Delta} \quad (1)$$

where Δ is a predefined value in $\left\{1/(p-1), \ldots, 1 - 1/(p-1)\right\}$, and *p* is a user-defined dividing level of each parameter range. A number of elementary effects can then be computed for each parameter by randomly sampling the parameter space according a strategy proposed by Morris (1991) (see Saltelli et al. (2008) for details). The sample mean μ of the elementary effects indicates the sensitivity of a certain parameter, with a larger μ indicating higher sensitivity. To circumvent the problem of opposite signs of μ cancelling out, which is common in non-monotonic models, we used a revised version of μ, namely μ^*. This is the mean of the absolute value of the elementary effects, i.e. $|EE_i|$ (Campolongo et al., 2007).

For each plot, the EE method was applied to all the parameters listed in Table 1, corresponding to each model output. The simulation end time was changed from 5 to 50 years and the screening process was repeated at each time step. A bootstrap technique (Efron and Tibshirani, 1993) was used to resample (with replacement) annual sensitivity measures (46 set of samples) to get a representative sensitivity measure (i.e., the mean of the distribution produced by the bootstrap) for each parameter corresponding to the specific model output. Based on this, the least influential parameters were screened out and parameters which have higher elementary effects for each model output were retained for FAST analysis. To simplify the graphical representations in the Results, we retained the 12 most influential parameters for each model output while holding the other parameters at constant default values. We then applied the FAST method to analyse the sensitivity of the selected parameters.

2.4. The FAST method

The FAST method, first proposed by Cukier et al. (1973), is a classic global sensitivity analysis technique, and suitable for both monotonic and non-monotonic models. The basic idea is to build space-filling curves for parameters in a *k*-dimensional parameter space with distinct frequencies. A Fourier transformation is then applied to the model output, and a spectrum is calculated at specific frequencies (corresponding to the frequencies of the parameters) to quantify the variance contribution of certain parameters to the whole variance of the model output (Sobol', 1990; Saltelli et al., 1999; Xu and Gertner, 2011).

Consider a model $y=f(x_1, x_2, ..., x_k)$, where x_i ($i=1, 2, \ldots, k$) is a model parameter that is rescaled to a k-dimensional unit hypercube $\Omega^k = \left\{x|0 \le x_i \le 1, i = 1, \ldots, k\right\}$. To perform a Fourier decomposition of f as a function of x_i in order to assess the sensitivities of each parameter, a periodic search function is used to explore the parameter space Ω^k and defined as follows (Saltelli et al., 1999):

$$x_i(s_j) = G_i(\sin(\omega_i s_j)) \tag{2}$$

where i is the parameter index; s is an independent variable varying over $(-\pi, \pi)$,

$$s_j = -\pi + \frac{\pi}{N} + \frac{2\pi(j-1)}{N} \tag{3}$$

where $j=1, 2, \ldots, N$, and N is an odd number representing the sample size. For each x_i, a distinct and incommensurate frequency ω_i is selected up to a given order M (usually set to 4 or 6). For a uniformly distributed x_i, we adopted $x_i(s_j)=(1/2)+(1/\pi)\arcsin(\sin(\omega_i s_j))$ as per Eq. (2) (Saltelli et al., 1999).

In FAST, a minimum sample size $N = 2M \max\left\{\omega_i\right\}$ is required by the Fourier theory (Saltelli et al., 1999), and the $\max\{\omega_i\}$ increases with parameter number, k (Cukier et al., 1975). For parameter-rich ecological models, the computational cost will be too high. Tarantola et al. (2006) proposed a random balance design (RBD) scheme to reduce the computational demand. The core idea of the RBD is to sample all the parameters in Eq. (2) using the same frequency ω (for simplicity can be set to 1), and for each x_i, a sample of N points is created using a random permutation of s, namely $(s_1^i, s_2^i, \ldots, s_N^i)$. Then the model is executed on the N permuted parameter values:

$$y(s_j) = f(x_1(s_j^1), x_2(s_j^2), \ldots, x_k(s_j^k)) \tag{4}$$

To calculate the partial variance of x_i, $y(s_j)$ is re-ordered to make the corresponding $x_i(s_j^i)$ in increasing order. The model output can then be decomposed into a discrete form (Cukier et al., 1978; Saltelli et al., 1999; Xu and Gertner, 2007):

$$V = \frac{1}{2}\sum_{k=1}^{\frac{N-1}{2}}(A_k^2 + B_k^2) \tag{5}$$

and

$$A_k = \frac{2}{N}\sum_{j=1}^{N} f(s_j)\cos(s_j k) \tag{6}$$

$$B_k = \frac{2}{N}\sum_{j=1}^{N} f(s_j)\sin(s_j k) \tag{7}$$

where k is an arbitrary integer from 1 to $(N-1)/2$. Here, $\Lambda_k = (1/2)(A_k^2 + B_k^2)$ is defined as the Fourier spectrum, then the partial variance of x_i can be calculated as the sum of the spectrum at $\omega=1$ and its higher harmonics (up to $M=6$ in this study) as follows:

$$V_i = \sum_{p=1}^{M} \Lambda_{p\omega} \tag{8}$$

The ratio of V_i/V represents the partial variance contribution to the total model output variance, and is the estimate of the main effect (i.e. sensitivity index) of x_i on y.

A large number of samples of the parameter space are required to guarantee the convergence and reliability of the sensitivity rankings and indices. To determine an appropriate sample size for simulation, the convergence times of the selected parameters by the EE method were analysed. We used StandVol to demonstrate the convergence of the FAST method and guide the selection of the appropriate simulation sample size. The simulation end age was set to 50 years.

2.5. Time-dependent sensitivity analysis

We kept the other site-specific parameters as default, and only changed the end age in the simulation from 5 to 50 years using an annual time step beginning from 1930. In each model run, the FAST procedure was performed for the selected model outputs. Except the value of the main effect index, a parameter sensitivity ranking for each model output at each time step was also created. The correlation in parameter sensitivity rankings at each pair of time steps (i.e., yr. 5 vs. yr. 6, yr. 5 vs. yr. 7, ..., yr. 5 vs. yr.50) was then calculated for each model output using Savage scores (Iman and Conover, 1987). Compared with Spearman's correlation coefficient, Savage scores are more sensitive to the agreement among the top rankings, which usually have more practical importance, while Spearman's correlation gives all ranks equal weighting. In SA, it is nature that the parameters at the bottom of the rankings generally have little influences on the output. So, the disagreement among the influential parameters should be highlighted compared with the disagreement among the uninfluential ones.

2.6. Influence of climate and soil

We also analysed the influence of climate and soil texture on the time-dependent sensitivities of the selected parameters. To assess the impact of climate data, based on the baseline scenario S0 (historical climate data of Greenvale), we applied three feasible climate change scenarios, namely S1 (1 °C warmer and 5% dryer), S2 (2 °C warmer and 15% dryer) and S3 (4 °C warmer and 25% dryer) (Bryan et al., 2010). For soil texture, three additional soil texture types were analysed (loamy sand (LS), sandy clay loam (SCL), and silty clay loam (TCL)). By changing each of these factors separately, the FAST procedure was re-run at stand ages from 5 to 50 years as above and compared with the original Greenvale sensitivities. We did not consider the interrelationships between forcing data variables such as VPD, temperature, rainfall, and soil texture. The authors acknowledge that using artificial climate scenarios while keeping the VPD as default will inevitably introduce bias into the results.

3. Results

3.1. Screening sensitive parameters

Table 2 lists the 12 parameters (excluding *SoilTexture*) remaining after the screening process corresponding to the model outputs StandVol and WF, respectively. The sensitivity rankings of *FR* and *SoilTexture* were very similar, and both outputs were very sensitive to *FR*. Most of the sensitive parameters shared by both outputs had similar sensitivity rankings. However, the sensitivity rankings of some specific parameters were quite different and show close relationships with the corresponding outputs. For example, *fracBB1* and *rho1* are highly sensitive for StandVol but not for WF. Conversely, *nWS*, *gammaF1* and *tWaterMax* were more sensitive for WF than for StandVol.

3.2. Convergence of FAST

Initially, the total main effects of the 12 (Table 2, excluding *SoilTexture*) parameters decreased quickly, then gradually stabilized around 1 after about 2000 model runs. The main effects of

Table 2
Rankings of the screened 12 relative sensitive parameters (excluding *SoilTexture*) corresponding to StandVol and WF using the elementary effects method, respectively. Smaller ranking number indicates higher sensitivity.

Parameter	StandVol	WF
Soil fertility rate		
FR	1	2
Allometric relationships & partitioning		
pRn	11	10
pRx	–	11
nWS	6	1
pFS20	9	6
Litterfall		
gammaF1	–	5
Fertility effects		
fN0	13	12
Specific leaf area		
SLA1	10	8
Light interception & VPD attenuation		
k	8	9
Rainfall interception		
tWaterMax	7	3
Production and respiration		
alphaCx	3	4
Conductance		
MaxCond	4	7
Branch and bark fraction		
fracBB1	5	–
Basic density		
rho1	2	–
Soil texture group factor		
SoilTexture	12	13

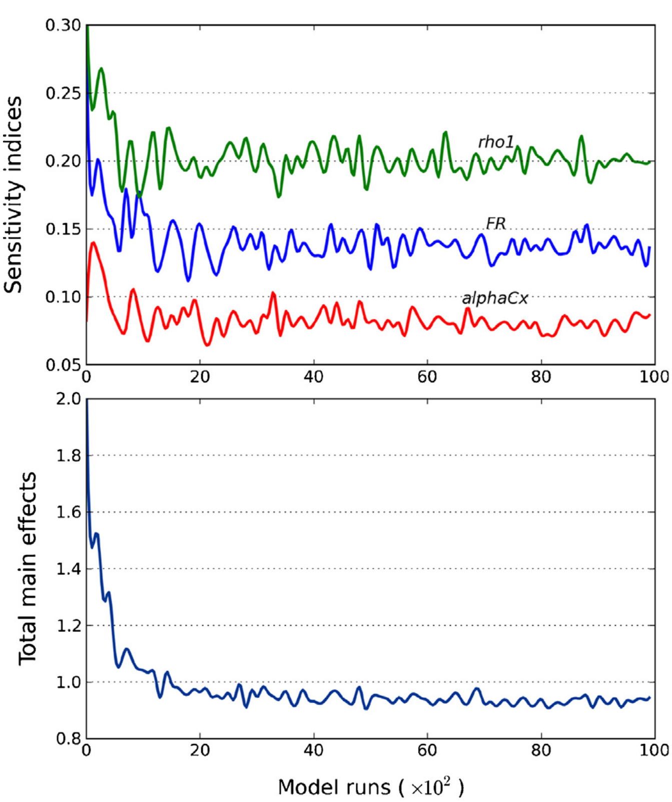

Fig. 1. An illustration about the convergence of the sensitivity analysis using FAST with the model output StandVol. Top – illustration of the evolution of the main effects of three selected individual parameters, i.e. (from top to bottom) *rho1* (maximum basic density for old trees), *FR* (fertility rating) and *alphaCx* (maximum canopy quantum efficiency). Bottom – the total main effects of the 12 parameters corresponding to StandVol.

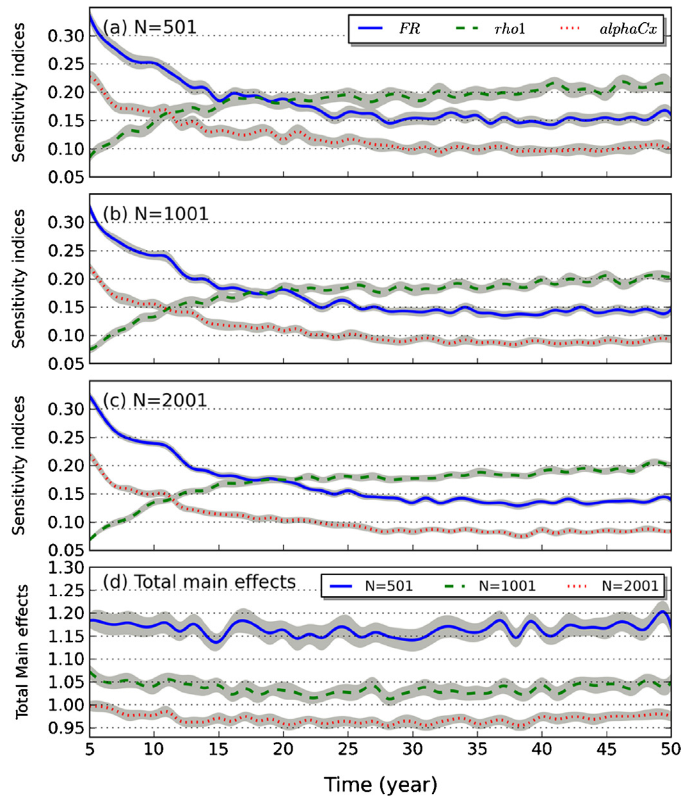

Fig. 2. Estimates of parameter sensitivity indices of *rho1*, *FR*, *alphaCx* (a, b and c) and the corresponding total main effects (d) from 1000 bootstrap replicates at sample sizes of N = 501, 1001 and 2001, including 95% confidence intervals (grey shade).

the selected parameters showed similar patterns (Fig. 1). According to FAST principles, the total main effects should equal 1 for an additive model, but for models in which non-monotonic processes and interactions between parameters exist, the total main effects should be less than 1 (Saltelli et al., 1999). This convergence guided selection of the sample size N = 2001 (N must be an odd number, see Section 2.4) for the following simulations. This sample size was also tested for WF, and it showed that convergence was also achieved. Larger sample sizes reduced both the uncertainty (thinner grey areas in Fig. 2) and the year-to-year variation in parameter sensitivities (smoother lines in Fig. 2). The uncertainty bounds on parameter sensitivities were fairly consistent over simulation time (Fig. 2).

3.3. Time-dependent parameter sensitivities

The sensitivities of most parameters changed substantially with stand age (Fig. 3). In general, three general sensitivity patterns were found in relation to stand age: increasing (e.g. *rho1* in StandVol and *gammaF1* in WF), decreasing (e.g. *alphaCx* and *k* in both outputs), and peaked (Λ-shaped) (e.g. *tWaterMax* and *SLA1* in WF). Only a few parameters, i.e. *fN0* and *pRx* in both outputs, displayed no obvious trends with stand age. These parameters typically had only minor influences on model outputs.

Savage scores revealed substantial differences in parameter sensitivity rankings at different stand ages, especially between the young (<10 year) and mature (>20 year) ages (Fig. 4). The greater the time differences between simulation periods, the lower the correlation in parameter sensitivity rankings for each output. For StandVol and WF, the sensitivity rankings were more similar at stand ages beyond 10 years.

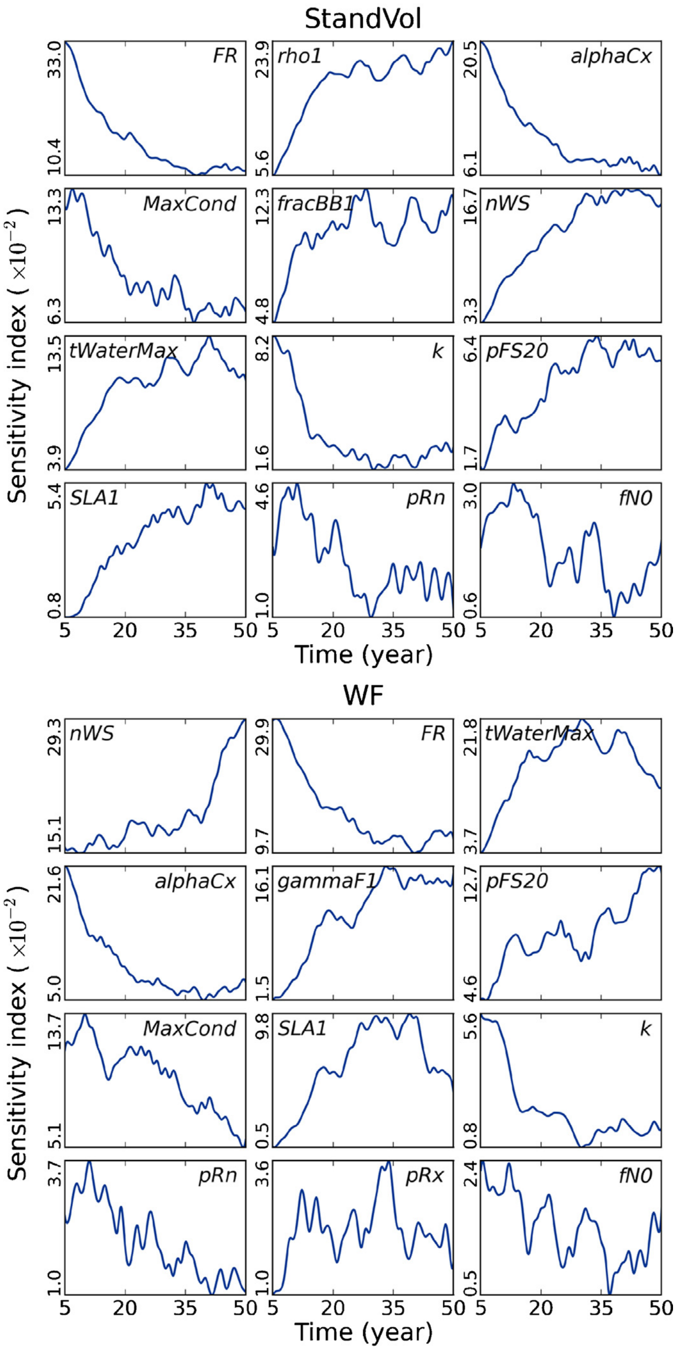

Fig. 3. Sensitivities of the 12 parameters for StandVol and WF over the 5–50 year simulations. The parameters for each model output are ordered according to their sensitivities derived from the EE method (Table 2) from high to low (from left to right, and then from top to bottom). To make the results more interpretable, a 5 point moving average was applied to each sensitivity data sequence.

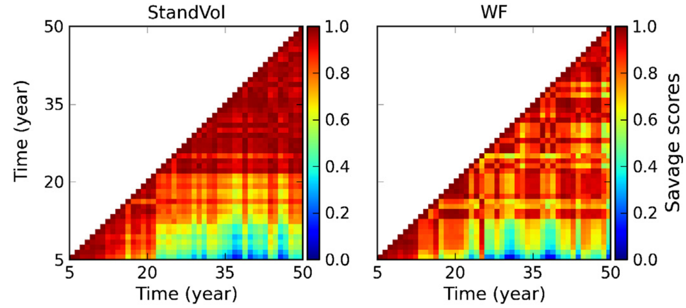

Fig. 4. Savage scores for parameter sensitivity rankings at different stand ages calculated based on simulations using default site-specific data.

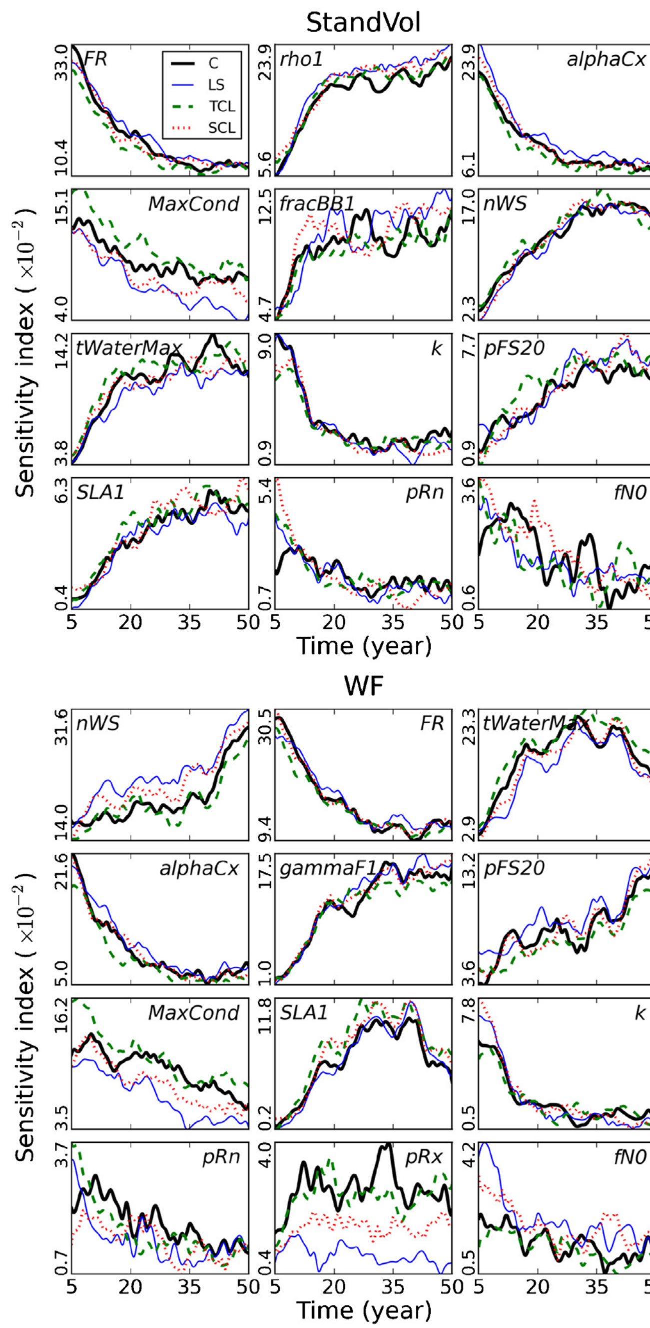

Fig. 5. Influence of soil texture on time-dependent parameter sensitivities. The black, blue, green and red lines represent clay (C), loamy sand (LS), silty clay loam (TCL) and sandy clay loam (SCL), respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

3.4. Influence of climate and soil texture

In general, changes in soil texture had less influence on parameter sensitivities than did changes in climate data. Whilst there were some localized fluctuations, soil texture had a relatively weak influence on parameter sensitivities in general, but the variations in *pRx* and *MaxCond* were noticeable (Fig. 5). However, substantial variations in the sensitivity of several parameters driven by the changes in climate data could be observed (Fig. 6). As the climate scenarios changed from S0 to S3 (becoming warmer and dryer), in general the

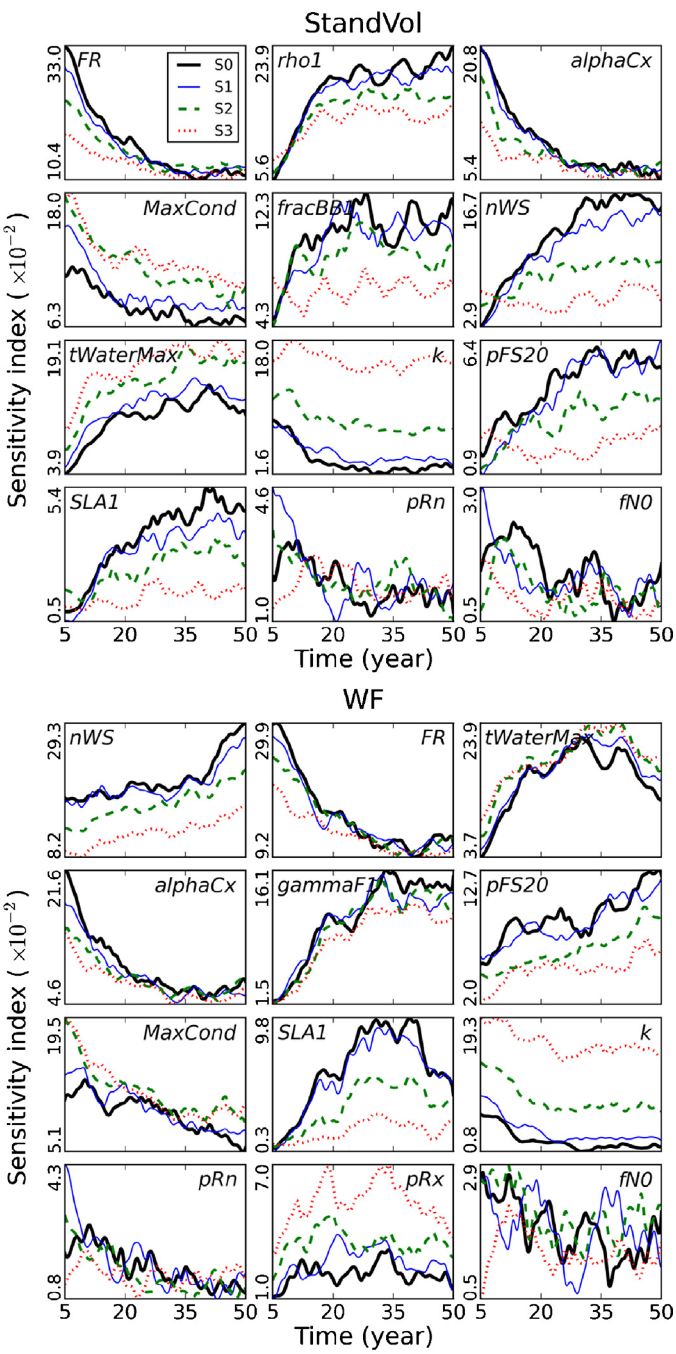

Fig. 6. Influence of climate scenarios on time-dependent parameter sensitivities. The black, blue, green and red lines represent climate scenarios S0, S1, S2 and S3, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

overall sensitivities of *MaxCond*, *k*, *tWaterMax* and *pRx* in StandVol and WF increased, whilst the sensitivities of the other parameters tended to decrease. These trends were more obvious in StandVol than WF. For *FR*, changing soil textures had little influence on its sensitivities (Fig. 5), while climatic warming and drying decreased parameter sensitivities, especially at young stand ages (Fig. 6). *pRn* and *fN0* showed no discernable sensitivity changes with stand age.

4. Discussion

We aimed to explore the time-dependence of parameter sensitivities in a temporal, process-based forest growth model. However, the basic principle applied in this study is not limited to 3-PG2, but is applicable for more general ecological models. The Morris method was effective in screening out the less influential parameters. Our application of the FAST method to 3-PG2 successfully tracked changes in sensitivities of the most influential parameters with model simulation period. The sensitivities of most parameters changed substantially with stand age. We also explored the influence of climate data and soil texture on parameter sensitivities over time. Whilst previous local and global SAs of both 3-PG and 3-PG2 (Almeida et al., 2004, 2007a; Esprey et al., 2004; Song et al., 2012) have increased the understanding of the model, these are static snapshots. The time-dependent sensitivity analysis presented here has provided a far deeper insight into how those sensitivities change over time. Proper examination of the time-dependent sensitivity of parameters and the influence of forcing data can increase understanding of model behaviour over time, especially for ecological models in which internal procedures or parameters usually have significant physiological meaning. For process-based ecological models, snapshot SAs generalize valuable information about model behaviour over time, which we believe is helpful for informing model development, understanding, and calibration. Whilst a few studies have touched on this issue (Cariboni et al., 2007; Xu et al., 2009; Makler-Pick et al., 2011; Song et al., 2012), this is the first study to quantify this effect.

4.1. Time-dependent parameter sensitivity

For 3-PG2, the relationships between parameter sensitivities and stand ages are typically one of three main types – decreasing, increasing, and peaked (Λ-shaped). Examination of these relationships can reveal certain details about the internal model structure. The sensitivities of some parameters declined over longer simulation periods. For example, *alphaCx* was a very sensitive parameter at young stand ages but its sensitivity declined with age (Fig. 3). In 3-PG2, *alphaCx* is directly used in the calculation of GPP (Landsberg and Waring, 1997). GPP is calculated by multiplying *alphaCx* by the intercepted photosynthetically active radiation, canopy cover and a series of site-specific (e.g. climate, soil water, and nutrition) and age modifiers (generally can be seen as a decreasing function) (Friedl et al., 1995; Landsberg and Waring, 1997). As the increment of canopy cover gradually slows down as forests mature, the age modifier continuously decreases and so does the influence (sensitivity) of *alphaCx*. The sensitivities of some parameters increased with stand age as more biomass accumulated in various parts of trees over time (e.g. *gammaF1*). Taking WF as an example, *gammaF1* acts as a scale factor in 3-PG2 and represents the ratio of lost biomass in WF (lossWF = *gammaF1**WF). In turn, WF is re-calculated by adding the difference between the increment (incrWF) and lossWF (i.e. WF + =incrWF-lossWF). Thus, changes in the relative variation of WF caused by *gammaF1* gradually increase with age, as well as the sensitivities of *gammaF1*. Although less common, the sensitivities of some parameters tended to start low, increase in the initial growth stages, peak when trees are at their maximum growth rates, and decline as trees matured (e.g. *tWaterMax* and *SLA1* in WF) (Fig. 3). For *tWaterMax* and *SLA1*, the peaked patterns were similar following the canopy development over time.

The climate data and soil texture also affected parameter sensitivities. For *FR*, it is more sensitive to climate changes, especially at the young stand ages. We speculated that this was due to changes in water availability which limits tree growth (especially in the rapid growth stage at young ages) and biomass allocation in different tree components (Cairns et al., 1997), and then decreasing *FR*'s sensitivities to biomass related model outputs. By changing soil texture, the relatively large changes in the sensitivity of *pRx* and the conductance-related parameter *MaxCond*, explicitly reflect the influences of soil texture on the available soil water status and root biomass (Landsberg and Waring, 1997). For *pRx*, climatic warming and drying increased its sensitivities, as the biomass allocation to roots will increase when water availability is limited (Landsberg and Waring, 1997).

4.2. Practical implications for ecological modelling

The results of this study suggest that SA studies, even global SAs, conducted at a single simulation time may provide an incomplete representation of parameter sensitivity. It follows then that model calibration and verification activities based on this incomplete picture can be biased and misguided. Sensitivity ranking is commonly used in SAs to indicate the relative importance of each parameter on specific model outputs (Esprey et al., 2004; Yang, 2011; Song et al., 2012). These rankings are often used to guide efforts towards improving the estimation accuracy and calibration of the most sensitive model parameters affecting the desired model outputs. However, model outputs found to be insensitive to variation in certain parameters early in the simulation, may become highly sensitive later on, and vice versa. The strong influence of forcing data (e.g. temperature) on parameter sensitivities also means that parameter sensitivities will vary from place to place. When spatially heterogeneous forcing data is involved, as is common in ecological models where ecological processes are driven by biophysical conditions, parameter sensitivities may be highly localized. Hence, conducting sensitivity analyses at the different simulation period and ignoring the influence of forcing data could lead to the targeting of resources at inappropriate parameters. In practice, SA is frequently used to identify insensitive parameters which can be fixed within a relatively wide range with little consequence for model outputs. However, without consideration of the time-dependence of parameter sensitivities and the state of the forcing data, incorrect parameters may be identified and set to default values, which may unknowingly exert significant influences on model outputs.

In this study, we used a simplified scheme to illustrate the possible influence of climate data to parameter sensitivities over time by changing temperature and rainfall only. However, it is noteworthy that in practical cases, the variations of temperature or rainfall will inevitably change the VPD, one of the primary factors which may have direct influence on canopy conductance and biomass production. So, it is suggested that full consideration should be taken about the interactions among the key climate variables in regional simulations, if data are available.

This leads us to suggest a refinement of the rule of thumb for the calibration of more complex, time-dependent ecological models. For these models, resources should be allocated to improve the accuracy of those parameters that are more sensitive at the desired simulation period and under the specific environmental conditions of interest. For example, for the simulation of young stands as is required in estimation of bioenergy production (Bryan et al., 2010), more attention needs to be paid to the most sensitive parameters for the desired model outputs early in the growth cycle (e.g. *SLA1* in WF). Conversely, for simulating processes such as carbon sequestration and biodiversity conservation (Crossman et al., 2011; Paterson and Bryan, 2012), the parameters most sensitive for the desired outputs at later stand ages are required (e.g. *rho1* in StandVol and *nWS* in WF). We recommend that the influence of forcing data (e.g. climatic data) should also be assessed for a complete assessment of model sensitivity (Fuentes et al., 2006; Spadavecchia et al., 2011). A more comprehensive SA scheme is required which takes into account the dynamics of parameter sensitivity over time and the impact of forcing data for complex, time-dependent ecological models. New methods are required to jointly quantify the sensitivities and uncertainties of parameters to specific model outputs.

For model practitioners, the above time-dependent SA practices should be integrated into the life-cycle of model development and calibration processes. As demonstrated in Section 4.1, the consecutive sensitivity indices of a series of parameters corresponding to a specific model output constitute a synthetic view of the interrelated model processes, and provide a good opportunity to check the soundness of model behaviour from the perspective of the evolving parameter sensitivities, but not limited to the model outputs (or state variables) as usually. And it could facilitate locating the inappropriate model processes by the incorrect sensitivity evolvements of certain parameters, especially for those parameters which have explicit ecological or biophysical meanings.

4.3. Future directions

The distributions of parameter values may have significant influences on parameter sensitivities (Helton et al., 2006), as well as the overall sensitivities at different stand ages. Tightening and loosening a parameter's distribution, or changing its shape (probability density function, PDF), may change its corresponding sensitivities. In this study, we used a uniform distribution for all parameters. An advance on the sensitivity analysis in this study is possible by using the more nuanced specification of parameter distributions. For example, parameter distributions can be specified and updated as more observed data comes in using a Bayesian strategy (Moradkhani et al., 2005; van Oijen et al., 2011). The probability distributions of parameters are determined by the availability and quality of the observed data, and this in turn will determine the corresponding sensitivities of the parameters for certain model outputs. This is especially the case for ecological models in which the distributions of many parameters have no explicit biophysical limitations and are largely determined by the observed data or the environmental factors.

Further advances are required in the coupling of model sensitivity and uncertainty analyses. The sensitivity of a parameter quantifies its contribution to the variance (uncertainty) in specific model outputs. Thus, the uncertainty of one model output is mainly dependent upon the uncertainties of the most sensitive parameters and the interactions among them. A natural extension of this research would be to examine how changes in parameter sensitivities and distributions affect the uncertainties in model outputs. This can be useful in model calibration as resources can then be targeted at key model parameters for reducing model uncertainty quantitatively. Another challenge is how to quantify the influence of the site-specific data (e.g. climate data) to get a more objective and comparable model sensitivity characterization.

5. Conclusions

Sensitivity rankings and indices are the goal of most sensitivity analyses and can be used to explain the relative importance of each parameter to specific model outputs. However, in this work we used a forest growth model and showed that parameter sensitivities may exhibit remarkable changes over time. Our results suggest that for temporal process-based ecological models, these are specific to the model simulation period and, to a lesser extent, forcing data. Sensitivity metrics need to be accompanied by the specific simulation period at which point the sensitivity results are conducted and the site-specific data used. Sensitivity analysis needs to be implemented at the specific simulation period of the process of interest in order to better inform efforts at improving the accuracy of the most sensitive model parameters. New methods are also required to jointly quantify parameter sensitivities and model uncertainties more fully over time, and special attention needs to be paid to parameters' distributions as these may also influence the overall parameter sensitivities. Special attention also needs to be paid about how to quantify the influence of site-specific data to get a "true" sensitive representation of the model itself.

Acknowledgements

This work was jointly supported by the National Science Foundation of China (31270588 & 31200363) and the National Basic Research Program of China (2012CB955304). We are grateful for the support of the China Scholarship Council and CSIRO's Sustainable Agriculture Flagship. We would like to acknowledge the assistance and helpful comments from Dr. Nicholas Coops, and anonymous referees for their constructive comments which greatly improved the paper.

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